Deep Learning Architectures

Dense · Convolutional · Recurrent · Transformer

All learn a parameterized map

$$\hat{y} = F_\theta(x)$$

They differ by the structural assumption used to process data.

Dense

1

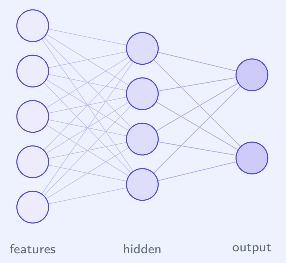
Dense architecture

Mathematical model

$$h^{(\ell)} = \phi\left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\right)$$

$W^{(\ell)} \in \mathbb{R}^{m_\ell \times m_{\ell-1}}$, $b^{(\ell)} \in \mathbb{R}^{m_\ell}$.

Principal layers



Inductive bias

Flat vector representation. Each output unit can combine every feature from the previous layer. Minimal structural prior.

Representative applications

tabular prediction · regression · risk scoring · baseline models · function approximation

Core limitation: ignores spatial or temporal structure unless engineered into the input.

Convolutional

2

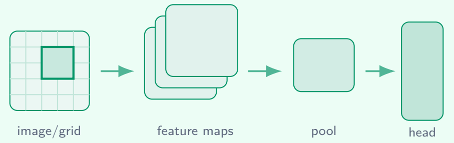
Convolutional architecture

Mathematical model

$$Y_{i,j,k} = \phi\left(b_k + \sum_{u,v,c} K_{u,v,c,k} X_{i+u,j+v,c}\right)$$

A shared filter K scans local neighborhoods of the input grid.

Principal layers



Inductive bias

Locality, parameter sharing, and translation equivariance. Nearby pixels or grid cells are assumed to form reusable patterns.

Representative applications

image classification · object detection · segmentation · medical imaging · spectrograms

Core limitation: global context usually requires depth, pooling, dilation, or attention.

Recurrent

3

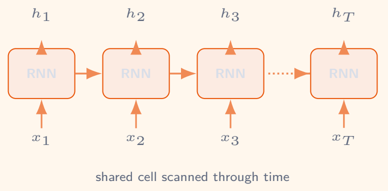
Recurrent architecture

Mathematical model

$$h_t = \phi(W_x x_t + W_h h_{t-1} + b_h), \qquad \hat{y}_t = g(W_y h_t + b_y)$$

LSTM/GRU variants add gates to control memory and gradients.

Principal layers



Inductive bias

Temporal order and stateful memory. The hidden state summarizes previous elements before processing the current input.

Representative applications

time series · speech · language modeling · biomedical signals · control

Core limitation: sequential dependency slows parallelization and may cause unstable gradients.

Transformer

4

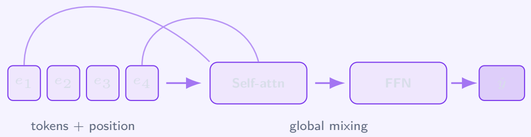
Transformer architecture

Mathematical model

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Principal layers



Inductive bias

Global pairwise interaction. Any token, patch, or element may attend to any other element in the input.

Representative applications

LLMs · translation · vision transformers · multimodal AI · protein modeling · code

Core limitation: vanilla self-attention scales as $O(n^2)$ with sequence length.

Comparison summary

Choose the architecture whose inductive bias matches the structure of the data.

Architecture	Best suited for	Main operation	Structural assumption	Main trade-off
Dense	Flat numerical or categorical feature vectors	Matrix multiplication + nonlinearity	All features may interact directly, but no spatial/temporal prior	Universal and simple, but parameter-heavy for structured data
CNN	Images, grids, spatial signals, spectrograms	Convolution + pooling	Local patterns repeat across positions	Efficient spatial feature extraction, but limited native long-range context
RNN	Ordered sequences and temporal signals	Recurrence through hidden state	Past elements influence future elements	Handles variable length, but training is sequential and less parallel
Transformer	Text, long sequences, images as patches, multimodal data	Scaled dot-product self-attention	Every element may be related to every other element	Powerful global context, but expensive for long inputs

Vector $\rightarrow$ Dense · Grid $\rightarrow$ CNN · Ordered stream $\rightarrow$ RNN · Global relations $\rightarrow$ Transformer